

ULIP Integration Requirement Document

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Revision History

Version	Date	Author	Reviewer	Significant Changes
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2.0	01/09/2021	Hasan Mhad Khan	Nayab Ali	Updated
3.0	29/11/2023	Ayush Arya	Nayab Ali	Added new field in 3 rd party response
4.0	07/12/2023	Ayush Arya	Nayab Ali	Added fois_02 new API
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Introduction

1.1 Purpose

The purpose of this document is to define and provide details for accessing API for Freight Operations Information System (FOIS) system information (such as tracking and monitor the movement of wagons, locomotives and unit trains) from ULIP system.

This document details the following points:

1. **Data Integration between FOIS and ULIP web service:** Identification of operational points where data integration between FOIS and ULIP web service will be needed.
2. **Technical Approach** for performing Data Integration

The Overall Description

Users will share information such as **FNR Number** based on data sent by user ULIP system will call respective **FOIS API** for accessing information.

Following are the points for accessing FOIS information: -

1. This API will share details based on FNR number.
2. ULIP will expose API for getting data from FOIS API.
3. **FOIS** API will take FNR number from user and get data from FOIS API.

1.2 Data Integration between FOIS System and ULIP application

User will share information to ULIP system based on data shared ULIP will provide data to user.

1.3 FOIS/01

This API will take the **FNR** number and connect with FOIS API to get data.
Request Attribute.

S. No.	Field Name	Field Description	Format	Required/Not – Required	Length
1	fnrnumber	FNR (freight name record) number for consignment tracking	[0-9]{11}	Required	11

1.3.1 Technical Approach

1. All the data will be shared through rest web services.
2. ULIP system will provide **FNR number** for which fois detail is required.
3. FOIS API will share details of fnr number which exist in FOIS system in JSON format.
4. ULIP system will provide the data in JSON Object format in response body.

Example: FNR number **21013108749** is shared by ULIP system, then ULIP system will make a request to FOIS API.

Request: The request must be of the following format, where fnr number is passed within the URL body in JSON format.

For production: <https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FOIS/01>

```
{"fnrnumber": "21030407355"}
```

Curl URL

```
curl --location --request POST 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FOIS/01' \
--
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ0ZXN0MSIsImhhdCI6MTYzM
DQ4NTY5MywiYXBwcyI6ImRhdGFwdXNoIn0.dWnnwihLmfcgjdErMlzNwyXTmJTAGuTE5w0Xq
05IPyhBISxThOvgHTCTNbtJg_PaGgg9bmWatdHkhOADN3xq3Q' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data-raw '{"fnrnumber": "00000023000"}'
```

For staging: <https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FOIS/01>

```
{"fnrnumber": " 23121326682"}
```

Curl URL

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FOIS/01' \
--header 'Authorization: Bearer
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGlwMiIsImhhdCI6MTY5MDAwMCwiYXBwcyI6ImRhdGFwdXNoIn0.2BrY8dCCS_BAH5Y6faXPYAPJEV52bGuZZRtKXD04AOmAUTOOBy9TC-
wmWVvxyAWsqLR9ifBhYD9xfqGMDPH9_A' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data '{"fnrnumber": "23121326682"}'
```

For obtaining authentication token, refer [Section 1.7](#)

Response:

```
{
  "response": [
    {
      "response": [
        {
          "newFnr": "",
          "etaDstn": "22:10 20-09-2023",
          "cmdt": "PHC",
          "currentStatus": "REACHED AT DESTINATION ON 22:10 20-09-2023",
          "lastRepStts": "AR",
          "locoNumb": "",
          "lastRepLocn": "M/S NABHA POWER LTD.SIDING(NPSB)",
          "lgtd": "76.515312",
          "fnrNo": "23091420258",
          "lttd": "30.538016",
          "stationFrom": "NEW KUSMUNDA COLLIERY SIDING, KORBA(NKCR)",
          "stationTo": "M/S NABHA POWER LTD.SIDING(NPSB)"
        }
      ],
      "responseStatus": "SUCCESS"
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

Status Code 200: FOIS data response

Status Code 400: Bad request (Invalid json syntax, invalid json key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server error occurred

Status Code 502: Server is not responding.

1.4 FOIS/02

This API will take the **sttnfrom**, **stnto**, **cmdt**, **wgontype** and connect with FOIS API to get data.

Request Attribute.

S. No.	Field Name	Field Description	Format	Required/Not – Required	Length
1	sttnfrom	Enter the station from as available in freight fois system	[0-9]{10}	Required	10

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2	stntno	Enter the station to as available in freight fois system	[0-9]{10}	Required	10
3	cmdt	Enter the cmdt as available in freight fois system	[0-9]{10}	Required	10
4	wgontype	Enter the wgon type as available in freight fois system	[0-9]{10}	Required	10

1.4.1 Technical Approach

1. All the data will be shared through rest web services.
2. ULIP system will provide **sttnfrom**, **stntno**, **cmdt** and **wgontype** for which fois detail is required.
3. FOIS API will share details of **sttnfrom**, **stntno**, **cmdt** and **wgontype** which exist in FOIS system in JSON format.
4. ULIP system will provide the data in JSON Object format in response body.

Request: The request must be of the following format, where sttnfrom, stntno, cmdt and wgontype is passed within the URL body in JSON format.

For production: <https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FOIS/02>

```
{ "sttnfrom": "NZM",
  "stntno": "HPP",
  "cmdt": "CEMENT",
  "wgontype": "BOXN"}
```

Curl URL

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FOIS/02' \
--header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiI1bGlwliwiaWF0IjoxNzI3NzYyNzcxLCJhcHBzIjoiaWoiZGF0YXB1c2gifQ.9_kJwxuXfJYomT0QeMgOSWi01EaxXQkZQ1JLevfNo-vJ5vjCY9olemZ_B4hiiq2ZCogyAbIUgL6YN-TVie3cA' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data '{ "sttnfrom": "NZM",
  "stntno": "HPP",
  "cmdt": "CEMENT",
  "wgontype": "BOXN"}
```

For staging: <https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FOIS/02>

```
{ "sttnfrom": "NZM",
```

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```
"stnto": "HPP",  
"cmdt": "CEMENT",  
"wgontype": "BOXN"}
```

Curl URL

```
curl --location --request POST 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FOIS/02' \  
-- \  
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWliOiJ1bGlwliwiaWF0IjoxNzAxOTMwNTI4LCJhcHBzIjoizGF0YXB1c2gifQ.N4WSBTuvN49BkpNKGBPIK1tv4npJtlucgU7remVqbERZn67LWSwvLYwHYabP9tQ7afFB8wxgpiv78QW-aSIWhw' \  
--header 'Accept: application/json' \  
--header 'Content-Type: application/json' \  
--header 'Cookie: __zlcmid=1DSISDwigKpTJa2; _fw_crm_v=b6a11a26-11c3-4b74-d0d1-3b9c43a5b7c7' \  
--data-raw '{ "sttnfrom": "NZM",  
  "stnto": "HPP",  
  "cmdt": "CEMENT",  
  "wgontype": "BOXN"}
```

For obtaining authentication token, refer [Section 1.7](#)

Response:

```
{  "response": [  
    {  
      "response": {  
        "stts": {  
          "evntCode": "FRGTDTLS",  
          "error": "",  
          "successFailFlag": "S"  
        },  
        "frgt": {  
          "tranloadcls": "140A",  
          "wgonothrchrg": "0.00",  
          "wgonfrgttaxwrds": "SIX LAKH SEVENTY-  
ONE THOUSAND EIGHT HUNDRED EIGHTEEN RUPEES ONLY",  
          "tranntr": "779.93",  
          "CCInfo": "CC+8",  
          "dist": "488",  
          "wgonbascfrgt": "746.10",  
          "transurchrg": "101.73",  
          "cmdtClsstl": "140A",  
          "trantotlfrgt": "5,84,072.40",
```

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```

"wgongsttax": "31,992.00",
"wgonrebtcont": "",
"rateWL": "746.10",
"surchrg": "DYNAMIC PRICING FOR COMMODITIES OTHER THAN COAL AND
COK",
"wgonchblwght": "680.0",
"wgonsurcont": "1",
"tranrebtmrk": "",
"rakeSize": "10",
"tranrebtamount": "",
"wgonothrcont": "3",
"tranfinlfrgt": "5,84,072.40",
"routeInfo": "Shortest",
"wgonfrgttax": "6,71,818",
"wgonloadfrgt": "940.92",
"PCC": "68.00",
"miniRake": "10",
"totlchblwght": "680.0",
"tranrebtchrg": "0.00",
"trangsttax": "29,204.00",
"tranothrchrg": "0.00",
"tranfullamount": "",
"wgonntr": "858.02",
"transurchrgdtls": [
  {
    "rebtamnt": "101.73",
    "rebtcode": "DPO",
    "prcn": "15.00",
    "desc": "DYNAMIC PRICING FOR COMMODITIES OTHER THAN COAL AN
D COKE(DPO)"
  }
],
"wgonrebtprcen": "",
"wgonsurchrg": "111.92",
"tranfrgttaxwrds": "SIX LAKH THIRTEEN THOUSAND TWO HUNDRED SEVENTY
-SEVEN RUPEES ONLY",
"rateTL": "678.20",
"tranbasfrgt": "678.20",
"tranothrchrgdtls": [
  {
    "rebtamnt": "39.00",
    "rebtcode": "DS",
    "prcn": "5.00",
    "desc": "DEVELOPMENT CHARGE(DS)"
  }
],
{

```

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```

        "rebtamnt": "20.00",
        "rebtcode": "OTC",
        "prcn": "0",
        "desc": "ORIGINATING TERMINAL CHARGES(OTC)"
    },
    {
        "rebtamnt": "20.00",
        "rebtcode": "DTC",
        "prcn": "0",
        "desc": "DESTINATION TERMINAL CHARGES(DTC)"
    }
],
"tranchrg": "79.00",
"tranloadfrgt": "858.93",
"tranrebtcont": "",
"wgonrebtamount": "",
"wgontotlfrgt": "6,39,825.60",
"atlrout": [
    {
        "sttno": "HARPALPUR- HPP (NC)",
        "sttnfrom": "HAZRAT NIZAM-UD-DIN JN.- NZM (NR)"
    }
],
"wgonothrrebt": "0.00",
"tranothrrebt": "0.00",
"wgonfullamount": "",
"othrchrg": "DEVELOPMENT CHARGE,ORIGINATING TERMINAL CHARGES,DE
STINATION TERMINAL CHARGE",
"tranfrgttax": "6,13,277",
"wgonrebtmrk": "",
"minmRake": "10",
"tranrebtprcn": "",
"wgonrebtchrg": "0.00",
"ETA": "20",
"cmdtClssWL": "140B",
"rebtchrg": "0",
"tranothrcont": "3",
"tranrebtchrgdtls": [],
"transurcont": "1",
"viaDesc": "TKD-AST-PWL-BTSR-MTJ-RKM-AGC-DHO-BLNR-GWL-VGLJ",
"wgonchrg": "82.90",
"wgonfinlfrgt": "6,39,825.60",
"wgonloadclss": "140B"
}
},
"responseStatus": "SUCCESS"

```

```

    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}

```

Status Code 200: FOIS data response

Status Code 400: Bad request (Invalid json syntax, invalid json key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server error occurred

Status Code 502: Server is not responding.

1.5 FOIS/04

This API will take the **Ingpwbltno** and connect with FOIS API to get data.

S. No.	Field Name	Field Description	Format	Required/Not – Required	Length
1	Ingpwbltno	Enter the Ingpwbltno as available in FOIS system.	[0-9]{10}	Required	10

1.5.1 Technical Approach

1. All the data will be shared through rest web services.
2. ULIP system will provide **Ingpwbltno** for which detail is required.
3. FOIS API will share details of Parcel tracking based on **Ingpwbltno** which exist in FOIS system in JSON format.
4. FOIS Integration will add JSON in **response** key in response body.
5. ULIP system will provide the data in JSON Object format in response body.

Request: The request must be of the following format, where **Ingpwbltno** is passed within the URL body in JSON format.

For production: <https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FOIS/04>

```

{"Ingpwbltno": "4002411032"}

```

Curl URL

```

curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FOIS/04' \
--header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJhcHBzIjoieXBpR2F0ZXdsInN1Yil6InVsaXAiLCJpYXQiOiJlE3NTc5OTMzMTR9.6pJ2ugsh7GodeNIKVJxR9wvPiWOJGt3qkCpHi5ob4fVbHihrMmrZh8hov0amY5kjiH7J7IGCc5_nFZ5JMHOW2Q' \
--header 'Accept: application/json' \

```

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```
--header 'Content-Type: application/json' \  
--data '{"Ingpwbldno": "4002411032"}'
```

For staging: <https://ulipstaging.dpiit.gov.in/ulip/v1.0.0/FOIS/04>

```
{'Ingpwbldno': "4002411032"}
```

Curl URL

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FOIS/04' \  
--header 'Authorization: Bearer  
eyJhbGciOiJIUzUxMiJ9.eyJhcHBzIjoieXBpR2F0ZXdheSIsInN1Yil6InVsaXAiLCJpYXQiOiJlE3NT  
c5MjQ5NzR9.AsGEMHb_74mog1lCzY60eK3gnCSpnCC_lpJB_-  
PiAlUa1axGFk0LMe8NpDA9bGnIPLBbPN37uzwA6sn4asvPQ' \  
--header 'Accept: application/json' \  
--header 'Content-Type: application/json' \  
--data '{"Ingpwbldno": "4002411032"}'
```

For obtaining authentication token, refer [Section 1.7](#)

Response:

To acknowledge the above request, ULIP will send the below response -

1. Invalid “Ingpwbldno” format

```
{  "response": null,  
    "error": "true",  
    "code": "400 BAD_REQUEST",  
    "message": "Data format failed OR wrong value entered at: Ingpwbldno. Format should follow  
[0-9]{10}"}
```

2. In case data do not exist for given “Ingpwbldno”

```
{  "response": [  
    {  
      "response": [  
        {  
          "error": "Invalid Receipt Number.."  
        }  
      ],  
      "responseStatus": "SUCCESS",  
      "message": null  
    }  
  ],  
  "error": "false",
```

```
"code": "200",  
"message": "Success"}
```

3. In case “Ingpwbltno” exist

```
{  
  "response": [  
    {  
      "response": [  
        {  
          "RR": {  
            "PWBLTNO": "4002411032",  
            "SCALE": "Luggage",  
            "BDT": "02-09-2025",  
            "CNEE": "RAM UDGAR",  
            "CNEEADD": "MIRZAPUR JYG",  
            "CNOR": "RAM UDGAR",  
            "FSTN": "BAREILLY JN.",  
            "ITMDET_PRNT": {  
              "ITMDET": {  
                "LOADSTNDET": "",  
                "UNLOADSTATUS_DT": "05-Sep-2025 00:52",  
                "LOADTRAIN": "14650",  
                "UNLOADSTNDET": "",  
                "DESC": "BICYCLE (40KG)",  
                "UNLOADTRAIN": "14650",  
                "STATUS_DT": "06-Sep-2025 08:26",  
                "STATUS": "Delivered",  
                "STNDET": "",  
                "TOTITMS": "2",  
                "LOADSTATUS_DT": "04-Sep-2025 03:37"  
              }  
            },  
            "CNORADD": "MIRZAPUR JYG",  
            "TSTN": "JYG-JAYNAGAR"  
          }  
        },  
        {  
          "responseStatus": "SUCCESS",  
          "message": null  
        }  
      ],  
      "error": "false",  
      "code": "200",  
      "message": "Success"  
    }  
  ]  
}
```

Status Code 200: FOIS data response

Status Code 400: Bad request (Invalid json syntax, invalid json key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server errors occurred

Status Code 502: Server is not responding.

1.6 Data Transmission & Authentication Mechanism

All data exchange would be done over secure HTTP (HTTPS). Request from ULIP's system would be made in FOIS system through a data exchange URL that has following form:

For production: <https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FOIS/01>

Body

```
{"fnrnumber":"21030407355"}
```

Access to above data exchange URL would be authorized by use of a security mechanism implemented by the ULIP system. It is as follows:

- First time access: It would require a username and password [which would be shared with user beforehand]. Once user hit with valid user id and password, ULIP system will return a basic authorization token to user, using that token user will communicate with ULIP system until token will not get expired.

Summary of HTTP Basic Authentication is as follows:

First time when User will hit ULIP secured API through username and password for access token as follow-

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/user/login' \  
--header 'Accept: application/json' \  
--header 'Content-Type: application/json' \  
--data-raw '{ "username": "xxxx", "password": "xxxx@123" }'
```

Authorization:

Bearer

```
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJpbmRlcmlvbnQpYXQjE10DU1NjYyMzc5ImFwcHMjOjJkYXRhcHVzaCJ9.jb-23Dbkwpz-3ptsuuZofzH-  
ce7jd0xR5WlkFfBPO_Dzjop01d5x90amqsjsqUARYLUPtUDiMX07oHG4L2AfffQ
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired, and user need to re authenticate.

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Using The above token, user can request ULIP system for FOIS details as follow -

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FOIS/01' \  
--header '  
'Authorization: Bearer  
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGwIiwiaWF0IjoxNzI3NzYyNzcwLCJhcHBzIjoizGF0  
YXB1c2gifQ.9_kJwxuXfJYoMt0QeMgOSWi01EaxXQkZQ1JLevfNo-  
vJ5vjCY9olemZ_B4hiiq2ZCog-yAbIUgL6YN-TVie3cA' \  
--header 'Accept: application/json' \  
--header 'Content-Type: application/json' \  
--data '{"fnrnumber": "23091420258"}'
```

For staging –

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/user/login' \  
--header 'Accept: application/json' \  
--header 'Content-Type: application/json' \  
--data-raw '{"username": "xxxx", "password": "xxxx@123"}'
```

Authorization:

Bearer

```
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGwIiwiaWF0IjoxNzI3MTYxMDgwLCJhcHBzIjoizGF0  
YXB1c2gifQ.XhVZm7NiCD05-  
v9091WJQhYzdk1a0Shs_Ley0T4R2MX63swRyxHbHxh6bVXtzqWvWiSLwwNGRQIcqGhr49hpGg
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired, and user need to re authenticate.

Using The above token, user can request ULIP system for details as follow -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FOIS/01>

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FOIS/01' \  
--header 'Authorization: Bearer  
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGwIiwiaWF0IjoxNzI3MTYxMDgwLCJhcHBzIjoizGF0  
YXB1c2gifQ.2BrY8dCCS_BAH5Y6faXPYAPJEV52bGuZZRtKXD04AOMAUtooBy9T  
C-wmWVvxyAWSqLR9ifBhYD9xfqGMDPH9_A' \  
--header 'Accept: application/json' \  
--header 'Content-Type: application/json' \  
--data '{"fnrnumber": "23121326682"}'
```